AMERICAN INTERNATIONAL UNIVERSTY-BANGLADESH

Faculty Of Science and Technology

Assignment Title: **Individual Assignment [Final]**

Course Name: Physics 2 [PHY1203]; Section: H

Course Teacher: MD. MASUD PARVEZ; Semester: Spring 2019-2020

Date Of Submission: 10 April 2020;

No	Name	ID	Program	Math
1.	HABIB, MD. IMTIAZ	19-39389-1	BSc CSE	

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1.	HABIB, MD. IMTIAZ	19-39389-1	BSc CSE	

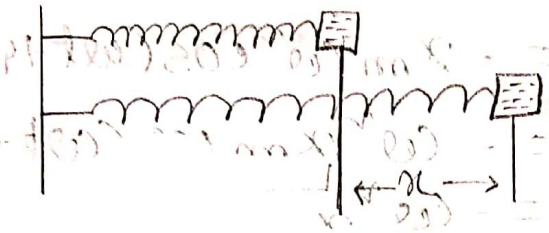
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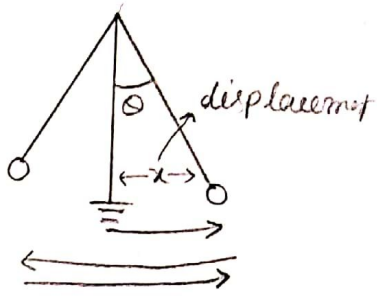
FINAL TERM

CHAPTER : 15 1st lecture: 09/03/20

1

Simple Harmonic Motion:





The equation of displacement of an object is in simple harmonic motion:

$$x = x_m \cos(\omega t + \phi)$$

- x: displacement
- x_m: max displacement
- ω: Angular frequency
- ϕ: Phase const

(i) Derive an expression of velocity & acceleration of an object is in simple harmonic motion:

→ velocity: $x = x_m \cos(\omega t + \phi)$

$$v = \frac{dx}{dt} = \frac{d}{dt}\left[x_m \cos(\omega t + \phi)\right]$$

$$= x_m \left\{-\sin(\omega t + \phi) \frac{d}{dt}(\omega t + \phi)\right\}$$

$$= -x_m \sin(\omega t + \phi) \cdot \omega$$

$$\therefore v = -x_m \omega \sin(\omega t + \phi) //$$

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→ Acceleration: $a = \frac{dv}{dt}$

$$= \frac{d}{dt}[-x_m \omega \sin(\omega t + \phi)]$$

$$= -x_m \omega \cos(\omega t + \phi) \cdot \omega$$

$$= -x_m \omega^2 \cos(\omega t + \phi)$$

$$\Rightarrow a = -\omega^2 x_m \cos(\omega t + \phi)$$

$$\therefore a = -\omega^2 x \quad //$$

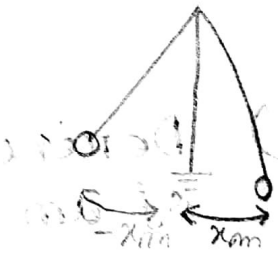
(ii) Find maximum displacement amplitude maximum velocity amplitude & maximum acceleration amplitude:

→ Maximeem displacement amplitude:

$$x = x_m \cos(\omega t + \phi)$$

$$\Rightarrow x = +x_m (\omega t + \phi) = 0°$$

$$\Rightarrow x = -x_m (\omega t + \phi) = 180°$$



$$\therefore x_{max} = \pm x_m \quad //$$

→ Maximum velocity amplitude:

$$V = -x_m \omega \sin(\omega t + \phi)$$

$$\Rightarrow V = -\omega x_m \quad [\because (\omega t + \phi = 90°)]$$

$$\& \quad V = +\omega x_m \quad [\because \omega t + \phi = 270°]$$

$$\therefore V = \pm \omega x_m \quad //$$

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→ Maximum acceleration amplitude:

$$a = -\omega^2 x$$

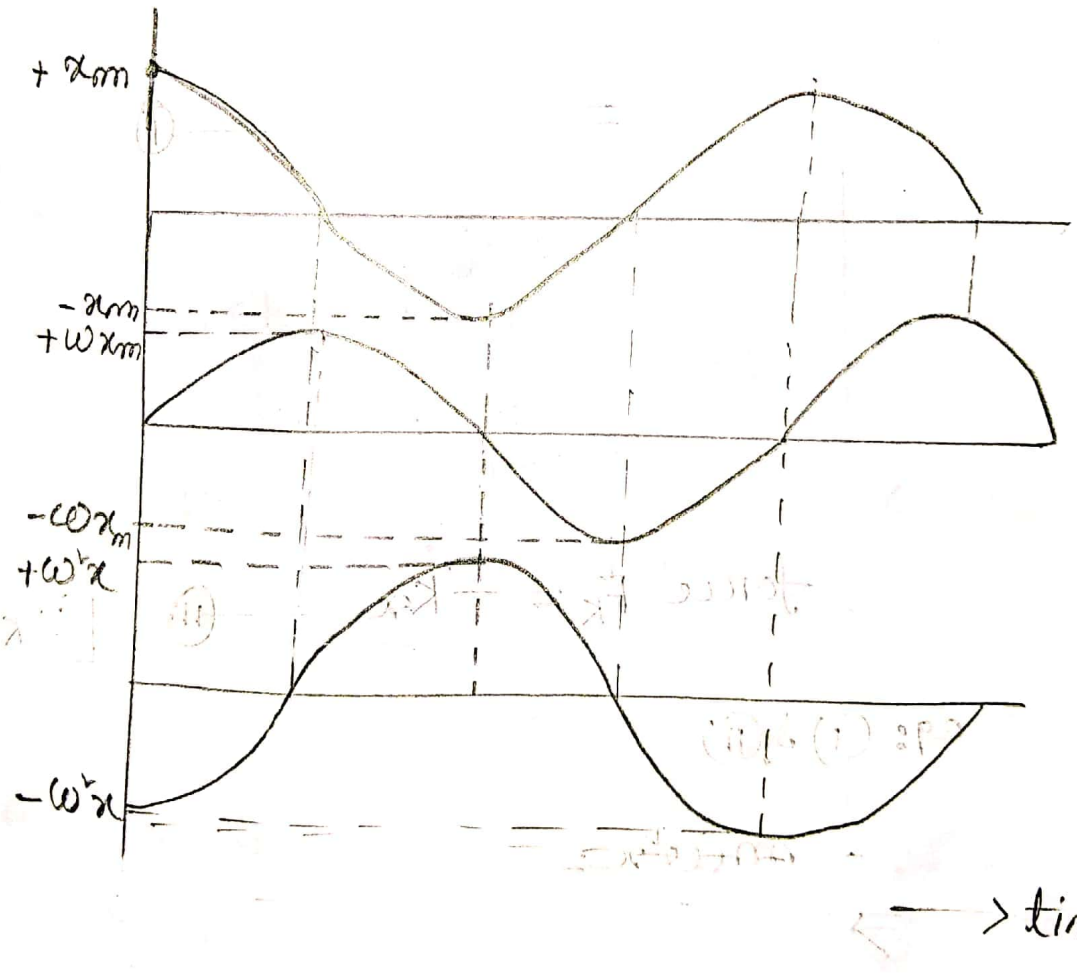
$$\Rightarrow a = -\omega^2 x_{max}$$

$$\Rightarrow a = -\omega^2 (\pm x_m)$$

$$\therefore a = \pm \omega^2 x_m //$$

(iii) Sketch the graphical representation of (x-t)(v-t)(a-t)

→



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14/03/20

2

For simple harmonic motion $x = x_m \cos(\omega t + \phi)$ show that the period of a simple harmonic oscillator, $T = 2\pi\sqrt{\frac{m}{K}}$?

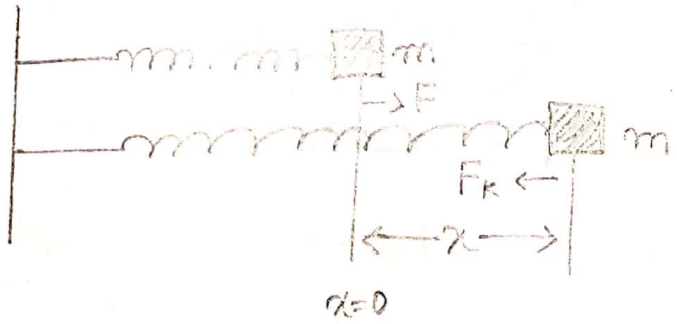
→ Applied force $F = ma$

For simple harmonic motion

$$a = -\omega^2 x$$

$$\therefore F = m(-\omega^2 x)$$

$$= -m\omega^2 x \quad \text{——(i)}$$



Spring force $F_K = -Kx$ ——(ii) $[\because K = \text{spring const}]$

From eq: (i) & (ii)

$$-m\omega^2 x = -Kx$$

$$\Rightarrow m\omega^2 x = Kx$$

$$\Rightarrow \omega^2 = \frac{K}{m} \Rightarrow \omega = \sqrt{\frac{K}{m}} \quad \text{——(iii)}$$

$\omega = 2\pi f$

$$\therefore 2\pi f = \sqrt{\frac{K}{m}}$$

$$\Rightarrow 2\pi \frac{1}{T} = \sqrt{\frac{K}{m}} \Rightarrow T = \frac{2\pi}{\sqrt{K/m}} = 2\pi\sqrt{\frac{m}{K}}$$

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2

$$a_{max} = \pm \omega^2 x_m$$

$$= \pm (2\pi f)^2 \times x_m$$

$$= \pm (2\pi \times 6.60)^2 \times 2.20 \times 10^{-2}$$

$$= \pm 37.83 \ ms^{-2}$$

#

(a) $T = 0.500 \ s$

(b) $f = \frac{1}{T} = \frac{1}{0.5} \ Hz$

$= 2 \ Hz$

(c) $\omega = 2\pi f = 2\pi \times 2$

$= 4\pi \ rads^{-1}$

(d) $K = m\omega^2$

$= 0.5 \times (4\pi)^2$

$= 78.96 \ Nm^{-1}$

(e) $V_{max} = \pm \omega x_m$

$= \pm (4\pi) \times 35 \times 10^{-2}$

$= 4.39 \ ms^{-1}$

(f) $|F_{max}| = m\omega^2 x_m$

$= 0.5 \times (4\pi)^2 \times 35 \times 10^{-2}$

$= 27.63 N$

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2nd Lecture:

3 The mechanical energy of a linear oscillator is indeed constant and independent of time:

$$E = U + K$$

$$= \frac{1}{2}Kx^2 + \frac{1}{2}mv^2$$

$$= \frac{1}{2}K\left[\{x_m\cos(\omega t+\phi)\}^2\right] + \frac{1}{2}m\{-\omega x_m \sin(\omega t+\phi)\}^2$$

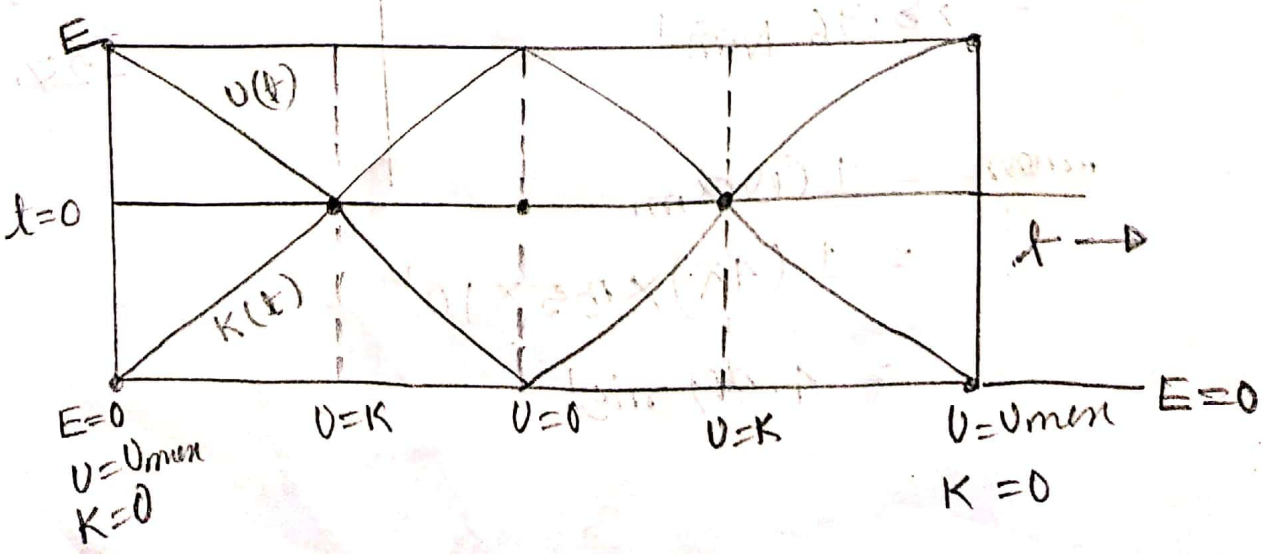
$$= \frac{1}{2}Kx_m^2\cos^2(\omega t+\phi) + \frac{1}{2}m\omega^2 x_m^2 \sin^2(\omega t+\phi$$

$$= \frac{1}{2}Kx_m^2\cos^2(\omega t+\phi) + \frac{1}{2}Kx_m^2 \sin^2(\omega t+\phi$$

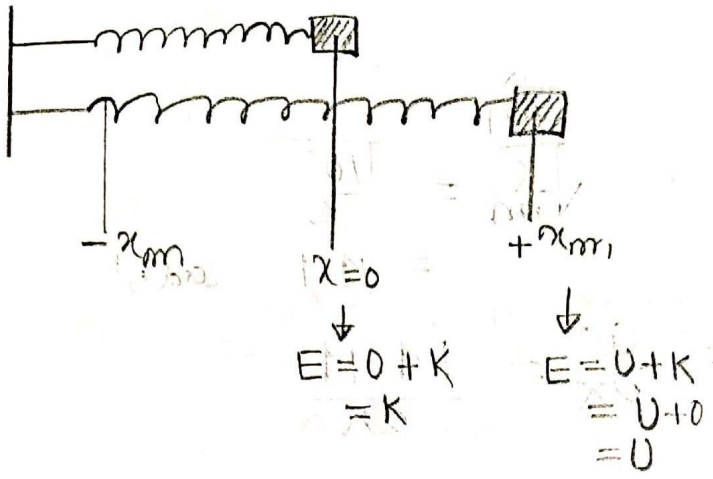
$$= \frac{1}{2}Kx_m^2\left(\sin^2(\omega t+\phi) + \cos^2(\omega t+\phi)\right)$$

$$= \frac{1}{2}Kx_m^2$$ //

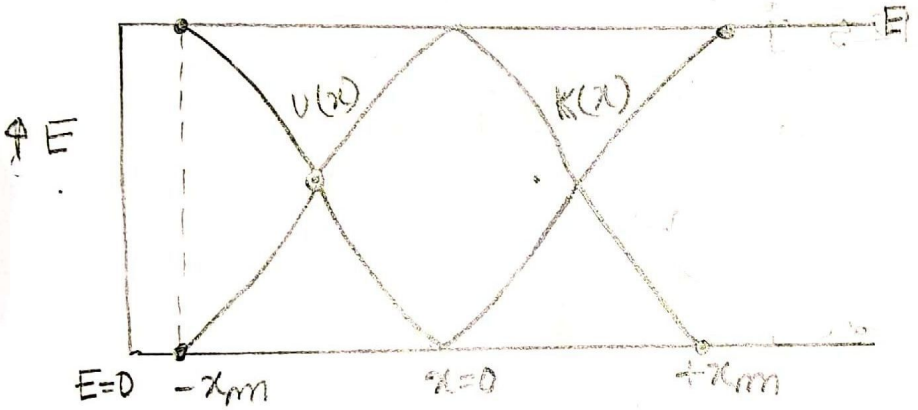
Sketch (U-t), (K-t), (E-t) graph:



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Sketch (U–x) (K–x) (E–x) graph:



#30 (a) $E = \frac{1}{2} K x_m^2$

$\Rightarrow K = \frac{2E}{x_m^2}$

$= \frac{2 \times 1}{(10 \times 10^{-2})^2}$

$= 200\ Nm^{-1}$

(b) $K = m\omega^2$

$\Rightarrow |V_{max}| = \omega x_m$

$\Rightarrow \omega = \frac{1.20}{10 \times 10^{-2}}$

$\therefore \omega = 12\ rads^{-1}$

$K = m\omega^2$

$\Rightarrow m = \frac{200}{(12)^2}$

$= 1.38\ kg$

(c) $\omega = 2\pi f$

$\Rightarrow f = \omega / 2\pi$

$= 12 / 2\pi = 1.9\ Hz$

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#31 (a) $K = m\omega^2$

$\Rightarrow \omega^2 = \frac{K}{m}$

$\Rightarrow \omega = \sqrt{\frac{K}{m}} = \sqrt{\frac{1000}{5}}$

$= 14.14 \text{ rad s}^{-1}$

$\therefore f = \frac{\omega}{2\pi} = \frac{14.14}{2\pi} = 2.25 \text{ Hz}$

(b) $U = \frac{1}{2} K x^2$

$= \frac{1}{2} \times 1000 \times (50 \times 10^{-2})$

$= 125 \text{ J}$

(c) $K = \frac{1}{2} m v^2$

$= \frac{1}{2} \times 5 \times (10)^2$

$= 250 \text{ J}$

(d) $E = 0 + K$

$\Rightarrow 375 = \frac{1}{2} K x_m^2$

$\Rightarrow x_m^2 = \frac{2 \times 375}{K}$

$\Rightarrow x_m = \sqrt{\frac{2 \times 375}{1000}}$

$= 0.87 \text{ m}$

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#36 At $t=0$, $\phi = \pi/6$

$$\frac{K.E}{U} = \frac{\frac{1}{2}mv^2}{\frac{1}{2}kx^2}$$

$$= \frac{\frac{1}{2}m[-\omega x_m \sin(\omega t+\phi)]^2}{\frac{1}{2}K[x_m \cos(\omega t+\phi)]^2}$$

$$= \frac{\frac{1}{2}m\omega^2 x_m^2 \sin^2(\omega t+\phi)}{\frac{1}{2}K x_m^2 \cos^2(\omega t+\phi)}$$

$$= \tan^2(\omega t+\phi)$$

$$= \tan^2(\omega \times 0 + \pi/6)$$

$$= \tan^2(\pi/6)$$

$$= \left(\frac{1}{\sqrt{3}}\right)^2 = \frac{1}{3} = 0.33 //$$

4

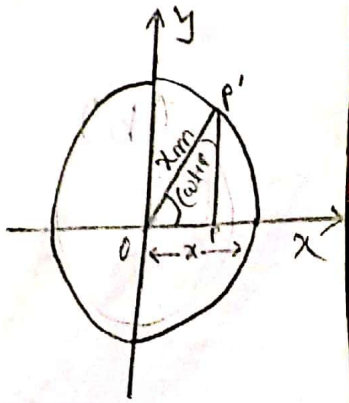
3rd Lecture:

Describe how simple harmonic motion is the projection of uniform circular motion on to a diameter of the circle:

→ For position or displacement:

$$\cos(\omega t+\phi) = \frac{x}{x_m}$$

$$\Rightarrow x = x_m \cos(\omega t+\phi)$$



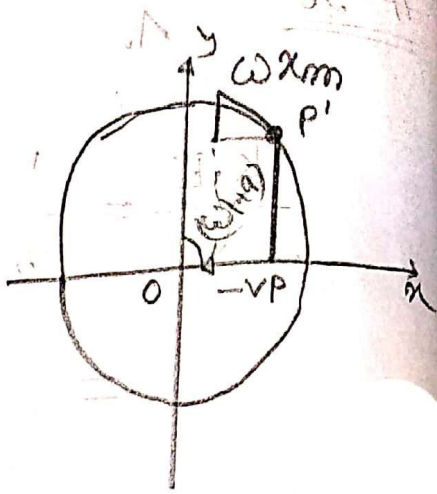
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→ For velocity:

$$\sin(\omega t+\phi) = \frac{-v}{\omega x_m}$$

$$\Rightarrow -v = \omega x_m \sin(\omega t+\phi)$$

$$\Rightarrow v = -\omega x_m \sin(\omega t+\phi)$$



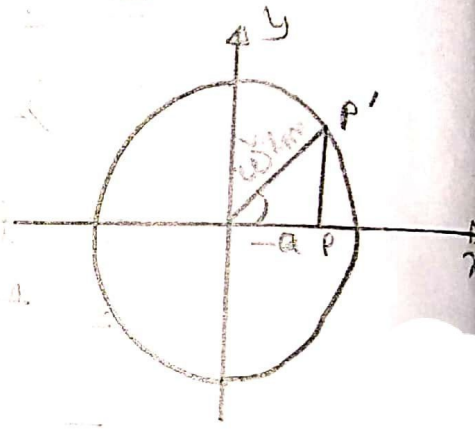
→ For acceleration:

$$\cos(\omega t+\phi) = \frac{-a}{\omega^2 x_m}$$

$$\Rightarrow -a = \omega^2 x_m \cos(\omega t+\phi)$$

$$\Rightarrow a = -\omega^2 x_m \cos(\omega t+\phi)$$

$$\therefore a = -\omega^2 x \;//$$



#77 (a) $K_{max} = \frac{1}{2}mv^2$

$$= \frac{1}{2}\times 20\times 10^{-3}(\omega x_m)^2$$

$$= \frac{1}{2}\times 20\times 10^{-3}\left(2\pi\frac{1}{T}\times x_m\right)^2$$

$$= 10\times 10^{-3}\left(2\pi\frac{1}{40\times 10^{-3}}\times \pi\times 10^{-2}\right)^2$$

$$\therefore K_{max} = 1.2\ \text{J} \;//$$

(b) $f = \frac{1}{T}$

$$= \frac{1}{40\times 10^{-3}}$$

$$= 25\ \text{Hz} \;//$$

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#78 (a) at $t = 40$ ms

$x = x_m$

$V = 0$

(b) $|a_{max}| = \omega^2 x_m$

$= (2\pi f)^2 \times x_m$

$= (2\pi \times 25)^2 \times 7 \times 10^{-2}$ [From 73]

$= 1727.18 \text{ ms}^{-2}$

1

4th Lecture: 16/03/20

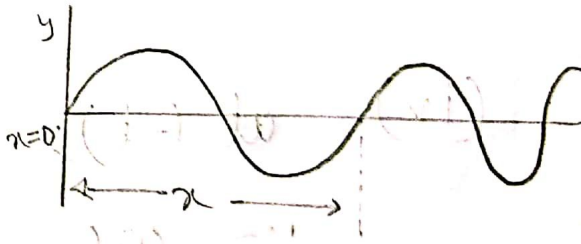
$y(x,t) = y_m \sin(kx - \omega t)$

(i) at $x = 0$;

$y(0,t) = \sin(-\omega t)$

at $x = \lambda$;

$y(\lambda,t) = \sin(k\lambda - \omega t)$



$\therefore k\lambda = 2\pi$

$\Rightarrow k = \frac{2\pi}{\lambda}$ //

(ii) at $t = 0$;

$y(x,0) = y_m \sin(kx)$

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at $t = T$;

$$y(x, T) = y_m \sin(kx - \omega T)$$

$\therefore \omega T = 2\pi$

$$\Rightarrow \omega = \frac{2\pi}{T} \quad //$$

(iii) at $(+t)$;

$$kx - \omega t = \text{const.}$$

$$\frac{d}{dt}(kx - \omega t) = 0$$

$$\Rightarrow k\frac{dx}{dt} - \omega\frac{dt}{dt} = 0$$

$$\Rightarrow kv - \omega \cdot 1 = 0$$

$$\Rightarrow v = \frac{+\omega}{k} \quad //$$

(iv) at $(-t)$;

$$kx - \omega(-t) = \text{const.}$$

$$\Rightarrow kx + \omega t = \text{const}$$

$$\Rightarrow \frac{d}{dt}(kx) + \frac{d}{dt}(\omega t) = 0$$

$$\Rightarrow kv + \omega = 0$$

$$\therefore v = -\frac{\omega}{k} \quad //$$

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5th Lecture:

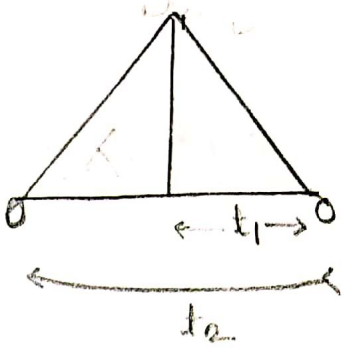
1

$y = +2;$

$\Rightarrow +2 = 6 \sin(kx + 600t_1 + \phi)$

$y = -2;$

$\Rightarrow -2 = 6\sin(kx + 600t_2 + \phi)$



$\sin(kx + 600t_1 + \phi) = \frac{1}{3}$

$\Rightarrow kx + 600t_1 + \phi = \sin^{-1}\left(\frac{1}{3}\right) = 0.34$ — (i)

$\sin(kx + 600t_2 + \phi) = -\frac{1}{3}$

$\Rightarrow kx + 600t_2 + \phi = \sin^{-1}\left(-\frac{1}{3}\right) = -0.34$ — (ii)

(ii) − (i) $\Rightarrow$ $600(t_2 - t_1) = -0.34 - 0.34$

$\Rightarrow t_2 - t_1 = \frac{-0.68}{600}$

$= 1.13 \times 10^{-3}$

$\therefore t_2 - t_1 = 0.0011\,s$ //

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#5

(a) The period $T = 4 \times t$

$= 4 \times 0.170$

$= 0.680\ s$

(b) The frequency of any wave is the reciprocal of its periodic time,

$$f = \frac{1}{T} = \frac{1}{0.68} = 1.47\ Hz$$

(c) The velocity of two wave is,

$$v = \lambda f$$

$$\therefore v = 1.4 \times 1.47 = 2.06\ m s^{-1}$$

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#7 (a) $\omega = \frac{v_m}{y_m} = \frac{16}{4\times10^{-2}} = 400 \text{ rad/s}$

$f = \frac{\omega}{2\pi} = \frac{400}{2\pi} = 63.66 \text{ Hz}$

(b) Given, $v = 80 \text{ m/s}$

$\therefore \lambda = \frac{v}{f}$

$= \frac{80}{63.66}$

$= 1.25 \text{ m/s}$

(c) Given equation, $y(x,t) = y_m \sin(kx \pm \omega t + \varphi)$

Substituting by $y = 4 \text{ cm}$ at $x = 0$, $t = 0$

$\therefore y_m = 0.04 \text{ m}$

(d) $k = \frac{2\pi}{\lambda}$

$= \frac{2\pi}{1.25} = 5 \text{ rad/s}$

(e) The angular velocity, $\omega = \frac{v_m}{y_m}$

$\Rightarrow \omega = \frac{16}{0.04}$

$\therefore \omega = 400 \text{ rad/s}$

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(f) $y = 4$ cm at $x = 0$, $t = 0$ therefore,

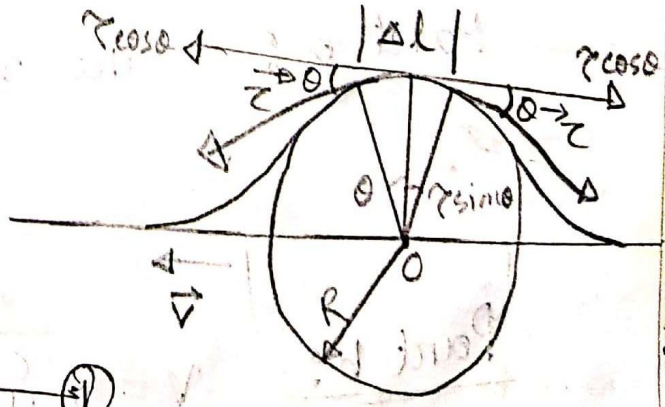
$$0.04 = 0.04 \sin(5(0) \pm 400(t) + \varphi)$$
$$\Rightarrow 1 = \sin(\varphi)$$
$$\Rightarrow \varphi = \sin^{-1}(1)$$
$$\therefore \varphi = \pi/2 \ //$$

$$\therefore Y = 0.04 \sin(5x \pm 400(t) + \tfrac{\pi}{2})$$

(g) The correct choice of sign in front of (t) is negative (−) because the wave is shown travelling to positive direction of x-axis.

2

Let's consider a small segment of length Δl and mass Δm,



$$F = \tau \sin\theta + \tau \sin\theta$$
$$= 2\tau \sin\theta$$
$$= 2\tau\theta \quad [\because \sin\theta = \theta]$$
$$= \tau(2\theta)$$
$$= \tau \frac{\Delta l}{R} \quad \left[\because \sin 2\theta = \frac{\Delta l}{R}, \ \therefore 2\theta = \frac{\Delta l}{R}\right] \text{ —— (i)}$$

$$\Delta m = \mu \Delta l \text{ —— (ii)}$$
$$a = \frac{v^2}{R} \text{ —— (iii)}$$

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$$\Rightarrow \frac{\tau \cancel{\Delta l}}{R} = (\mu \cancel{\Delta l}) \times \frac{v^2}{R}$$

$$\Rightarrow \frac{\tau}{\mu} = v^2$$

$$\Rightarrow v = \sqrt{\frac{\tau}{\mu}}$$ //

CHAPTER: 16 6TH Lecture:

#14 Given $y = 2m \left[\sin (20m^{-1})x - (600s^{-1})t\right]$

$\tau = 15N$

Comparing the given equation with the general wave function form:

$$y = y_m \sin(kx - \omega t + \phi)$$

$$k = \frac{2\pi}{\lambda} = 20 \text{ rad/m}$$

$$\omega = 2\pi f = 600 \text{ rad/s}$$

Part a: The wave speed, $v = \frac{\omega}{K}$

$$= \frac{600}{20}$$

$$= 30 \text{ m/s}$$

Part b: $v = \sqrt{\frac{\tau}{\mu}}$

$$\Rightarrow v^2 = \frac{\tau}{\mu} \Rightarrow \mu = \frac{15}{(30)^2} = 17 \text{ g/m}$$

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3. Wave functions of two sinusoidal transverse waves travelling along ——— find their resultant wave and sketch a plot of the wave:

→

$$y_1(x,t) = y_m \sin(kx - \omega t)$$
$$y_2(x,t) = y_m \sin(kx - \omega t + \phi)$$

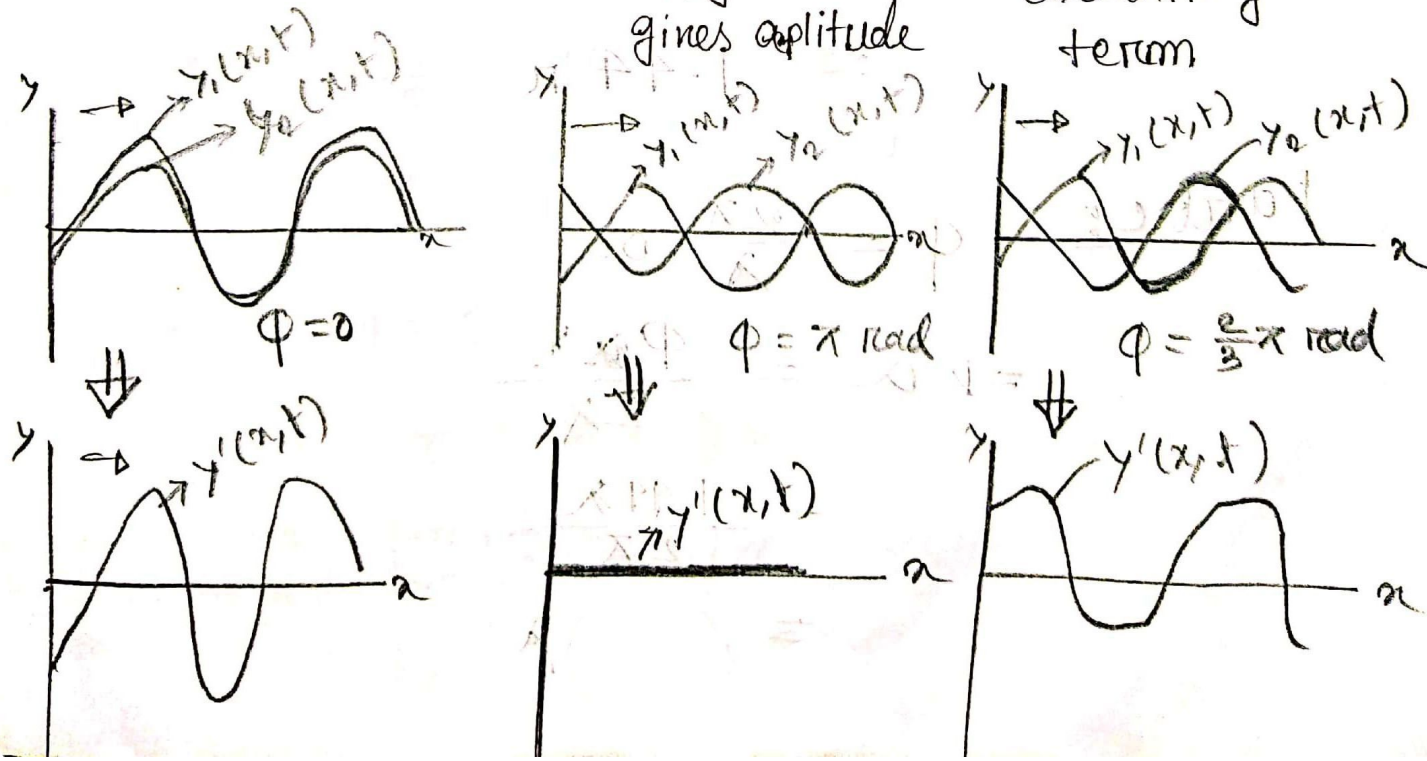
$$y'(x,t) = y_1(x,t) + y_2(x,t)$$
$$= y_m \sin(kx - \omega t) + y_m \sin(kx - \omega t + \phi)$$

Using $\sin\alpha + \sin\beta = 2\sin\frac{1}{2}(\alpha+\beta)\cos\frac{1}{2}(\alpha-\beta)$

$$y'(x,t) = \left[2 y_m \cos\tfrac{1}{2}\phi\right] \sin\left(kx - \omega t + \tfrac{1}{2}\phi\right)$$

Magnitude gives amplitude

Oscillating term



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#32 By using formula

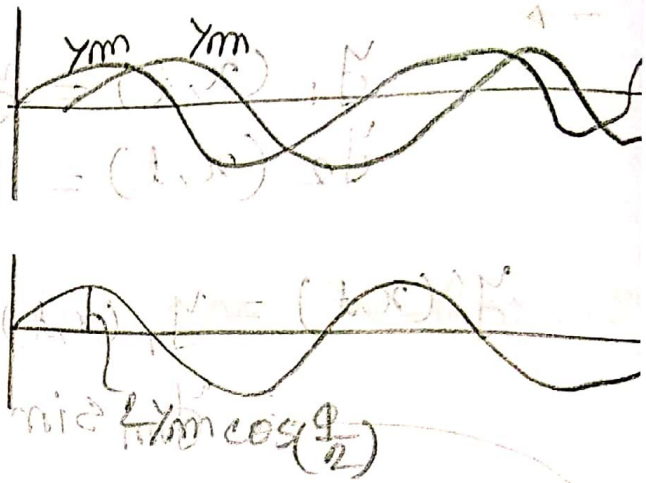
$$Y(x,t) = 2y_m \cos\left(\frac{\phi}{2}\right) \sin(kx - \omega t + \phi)$$

By comparing with the general form

$$2y_m \cos\left(\frac{\phi}{2}\right) = 1.5\, y_m$$

$$\Rightarrow \frac{\phi}{2} = \cos^{-1}\left(\frac{1.5}{2}\right)$$

$$\Rightarrow \frac{\phi}{2} = 41.4^\circ$$



Part a: $\phi = 2 \times 41.4^\circ$

$= 82.8^\circ$

Part b: $\phi = \frac{82.8\pi}{180}$

$= 1.44$ rad

Part c: $\phi = \frac{2\pi}{\lambda} a$

$$\Rightarrow a = \frac{\phi \lambda}{2\pi}$$

$$= \frac{1.44\lambda}{2\pi}$$

$$= 0.23\lambda$$

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7th Lecture:

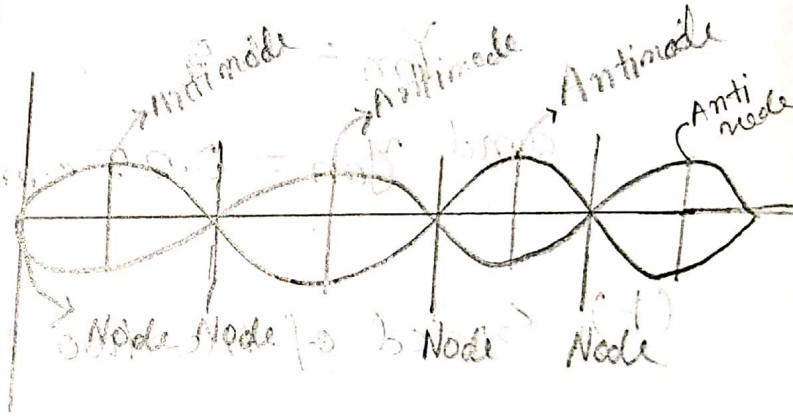
4. Two sinusoidal wave travelling in the opposite direction along a stretched string given by $y_1(x,t) = y_m \sin(kx - \omega t)$, $y_2(x,t) = y_m \sin(kx + \omega t)$ — Find the position of nodes and antinodes & sketch the figure when the waves are in phase of out of phase.

⟹ $y_1(x,t) = y_m \sin(kx - \omega t)$

$y_2(x,t) = y_m \sin(kx + \omega t)$

∴ Combined wave,

$y(x,t) = [2 y_m \sin kx] \cos \omega t$



For nodes:

$\sin kx = 0$

$\Rightarrow kx = n\pi \quad [n = 0, 1, 2 \ldots]$

$\Rightarrow x = n\frac{\lambda}{2} \quad \left[\because k = \frac{2\pi}{\lambda}\right]$

Antinodes:

$|\sin kx| = 1$

$kx = \frac{1}{2}\pi, \frac{3}{2}\pi, \frac{5}{2}\pi$

$\Rightarrow kx = (n + \frac{1}{2})\pi \quad [n = 0, 1, 2 \ldots]$

$\Rightarrow x = (n + \frac{1}{2})\frac{\lambda}{2} \quad \left[\because k = \frac{2\pi}{\lambda}\right]$

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#53

By comparing the given the equation

$$K = \frac{2\pi}{\lambda} = \frac{\pi}{3} \text{ rad/cm}$$

$$\omega = 2\pi f = 40\pi \text{ rad/s}$$

(a) comparing with the general equation:

$$y_m = 0.5 \text{ cm}$$

and $y_m = 0.25$ cm

(b) Speed of the two waves, $v = \frac{\omega}{K}$

$$= \frac{40\pi}{\frac{\pi}{3}}$$

$$= 120 \text{ cm/s}$$

(c) Distance between each two consecutive nodes:

$$K = \frac{2\pi}{\lambda}$$

$$\Rightarrow \lambda = \frac{2\pi}{K} = 6$$

$$\therefore D = \frac{6}{2} = 3 \text{ cm}$$

(d) The transverse speed of a particle on the string is

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$$u = \frac{dy}{dt} = -\omega y_m \sin(kx)\sin(\omega t)$$

at $x = 1.5\,cm$ and $t = \frac{9}{8}\,s$

$$\Rightarrow u = \frac{dy}{dt} = -40\pi y_m \sin\left[0.015\times\left(\frac{\pi}{3}\right)\right]\sin\left[40\pi\left(\frac{9}{8}\right)t\right]$$

$$= 0$$

#76 A standing wave results from the sum of two transverse travelling waves given by $y_1 = 0.050\cos(\pi x - 4\pi t)$ and $y_2 = 0.050\cos(\pi x + 4\pi t)$ ——— (a) what is the smallest positive value of x that corresponds to a node? (b) First (c) second & (d) third time of the particle at $x=0$ has zero velocity.

→ We know, $y'(x,t) = 2y_m \sin(kx)\cos(\omega t)$

But the given equation contains only cosine

$\therefore y'(x,t) = 2y_m \cos(kx)\cos(\omega t)$

By substituting, $y'(x,t) = 0.1\cos(\pi x)\cos(4\pi t)$

For node, $\cos\pi x = 0$

Solving for x, $\pi x = n\frac{\pi}{2}$ $[n = 0, 1, 2, 3 ---]$

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$\Rightarrow x = \frac{n}{2}$

∴ The smallest position value of x that corresponds to a node is $n=1$, where $x = \frac{1}{2}m$ //

(b) The velocity is given by,

$$y' = 2 \times 0.05 \cos(\pi x) \cos(4\pi t)$$

$$\frac{dy'}{dt} = -4\pi \times 0.1 \cos(\pi x) \sin(4\pi t)$$

Substituting $x=0$, when the velocity vanishes we have,

$$\sin(4\pi t) = 0$$

Solving for t we have

$$4\pi t = n\pi, \text{ where } n = 0, 1, 2, 3, \ldots$$

Therefore, $t = \frac{n}{4}$

The first time at which the particle has zero velocity is at $n=0$, where $t=0$ //

(c) For the second time, at $n = 1$

$t = \frac{n}{4} = \frac{1}{4}$ s //

(d) For the third time, at $n=2$

$t = \frac{n}{4} = \frac{1}{2}$ s //

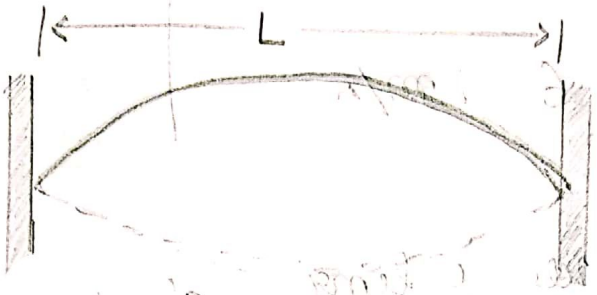
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8th Lecture:

5 ⊞ Derive expressions for resonant frequency of a stretched string of length L with ends:

→

a.



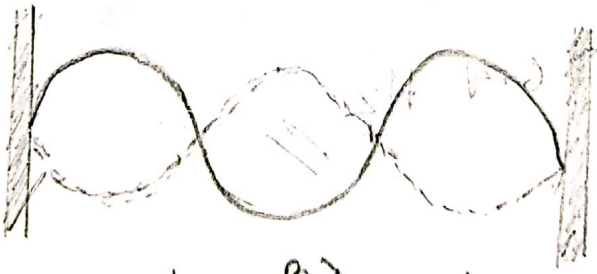
$L = \frac{\lambda}{2}$, for single loop

b.



$L = \frac{2\lambda}{2}$, for two loops.

c.



$L = \frac{3\lambda}{2}$, for three loops

$L = n\frac{\lambda}{2}$

$\Rightarrow \lambda = \frac{2L}{n}$ for $(n = 1, 2, 3 \text{---})$

$V = f\lambda$

$\Rightarrow f = \frac{V}{\lambda} = n\frac{V}{2L}$ for $n = 1, 2, 3 \text{---}$

$\therefore f = n\frac{V}{2L}$ //

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#44 (a) $v = \sqrt{\frac{T}{\mu}}$

$= \sqrt{\frac{7}{2\times10^{-3}/1.25}}$

$= 66.1 \text{ m/s}$

$\mu = \frac{2\times10^{-3}}{1.25} = \frac{m}{L}$

(b) The wave length of the wave with the lowest resonant frequency f_1 is $\lambda_1 = 2L$

where, $L = 125 \text{ cm}$

So,

$f_1 = n \frac{v}{2L}$

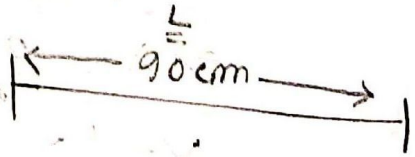
$= \frac{66.1}{2\times1.25} \times 1$

$= 26.4 \text{ Hz}$ //

#49 (a) $v = \sqrt{\frac{T}{\mu}}$

$= \sqrt{\frac{150}{7.20\times10^{-3}}}$

$= 144.33 \text{ m/s}$



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(b)

wave length

$\lambda = ?$

$n = 3$



$n = 3$

$$L = n \frac{\lambda}{2}$$

$$\Rightarrow \lambda = \frac{2L}{n}$$

$$= \frac{2 \times 0.9}{3}$$

$$= 0.6 \text{ m}$$

(c) We know,

$$v = f\lambda$$

$$\Rightarrow f = \frac{v}{\lambda}$$

$$= \frac{144.33}{0.6}$$

$$= 240.55 \text{ Hz} \quad //$$

Note: $f = n \frac{v}{2L}$; For resonant frequency.

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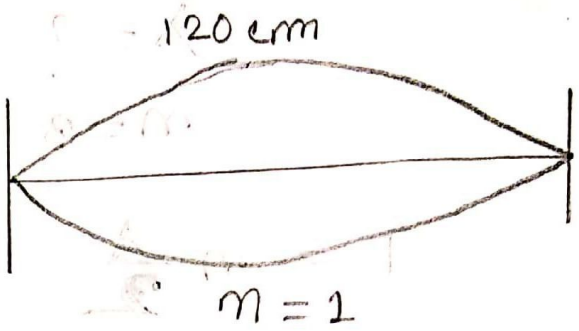
#85

(a) $L = n\frac{\lambda}{2}$

$\Rightarrow \lambda = \frac{2L}{n}$

$= \frac{2 \times 1.2}{1}$

$= 2.4\ m$



$n = 1$

(b) $\lambda = \frac{2 \times 1.2}{2}$

For second longest $\lambda = 1.2\ m$

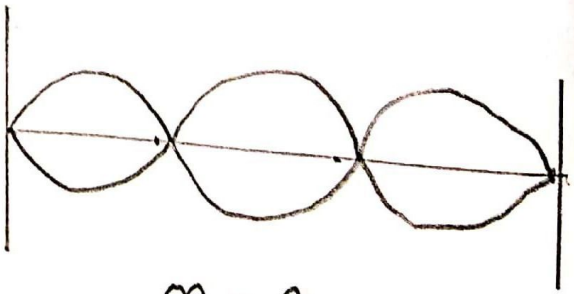


$n = 2$

(c) For the third longest

$\lambda = \frac{2 \times 1.2}{3}$

$= 0.8\ m$



$n = 3$

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